작성 안내 (제출 전 이 페이지 삭제)

이 초안의 구조. 결과와 무관한 부분(서론·이론, 실험 방법, 고찰의 이론적 틀, 참고문헌)은 완성된 문장으로 써 두었습니다. 실험 후 바뀌는 값과 관찰은 회색 상자로 남겼습니다. 상자 안의 숫자는 매뉴얼의 계획값이므로 **반드시 본인 측정값으로 바꿔야 합니다**. 결과에 따라 문장이 달라지는 곳은.tex 파일에 % 주석으로 선택지를 달아 두었습니다.

실험 후 채울 것

1. 2절과 표 1: 실제 질량, KOH 방울 수, 추가 styrene 여부, 재결정에 더 넣은 에틸아세테이트 양.
2. 3절: TLC 거리와 R_f(표 3, datasheet 그림 옮기기), 생성물 질량·수율, 녹는점, IR, ^{1}H NMR의 비방향족 H 세 개(δ, J), X선 구조로 확정한 생성물 번호.
3. 수율 계산 기준: X선 결과가 **3A/3B**이면 MW 488.63, **4**이면 460.62. 표 2와 초록의 해당 줄만 남깁니다.
4. 4.3–4.4절: 주석의 선택지 중 결과에 맞는 문장 선택.

제출 전 확인할 것 [CONFIRM]

1. 유기 파트 보고서 형식·분량·마감(평가 방식 공지 확인). proforma 형식이면 문단을 답안으로 옮겨 쓰면 됩니다.
2. 과목의 생성형 AI 사용 규정(사용 사실 표기 필요 여부).
3. NMR 측정 주파수(400 MHz로 가정), IR 기기명, 녹는점 측정기.
4. 강의에서 제공될 X선 구조 자료를 그림으로 넣을지.

컴파일: XeLaTeX. 파일 첫 줄 근처의 `\drafttrue`를 `\draftfalse`로 바꾸면 이 페이지가 빠집니다.

One-Pot Aldol Condensation and Diels–Alder Reaction: Tetraphenylcyclopentadienone and Its Cycloaddition with Styrene

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CM3191 Chemical Synthesis Experiments, Organic Experiment 2

Thursday laboratory, fume hood B7 • Date of experiment: 8 October 2026

Instructor: Dr Hoang Truong Giang

Abstract. 2,3,4,5-Tetraphenylcyclopentadienone (tetracyclone, **1**) was generated in situ by the base-catalysed double aldol condensation of benzil with 1,3-diphenylacetone in triethylene glycol, and was trapped in the same vessel by a Diels–Alder reaction with styrene at 110 °C. The recrystallised product was obtained as a [white] solid ([0.XXX] g, [XX]%) with mp [XXX–XXX] °C. On the basis of IR and ^{1}H NMR data and the X-ray crystal structure, it was identified as [3A / 3B / 4].

1. Introduction

The aldol condensation and the Diels–Alder reaction are two of the most widely used carbon–carbon bond-forming reactions. Both are atom-economical,[11] and in this experiment they are combined in one pot: the intermediate diene is not isolated but reacts directly with the dienophile. This saves the solvent and materials needed to isolate the intermediate, reduces the number of vessels and allows a reactive intermediate to be trapped in situ.[1]

Aldol condensation. In the presence of KOH, 1,3-diphenylacetone is deprotonated at a carbon that is activated by both the carbonyl group and a phenyl ring. The resulting enolate adds to one carbonyl group of benzil, and the aldol adduct dehydrates to an enone. A second, intramolecular aldol addition followed by dehydration closes the five-membered ring and gives tetracyclone **1** (Figure 1), a classic preparation.[2] The loss of two molecules of water drives the reaction forward. Compound **1** is deep purple because of its extended conjugation, which makes its formation easy to follow by eye and by TLC.

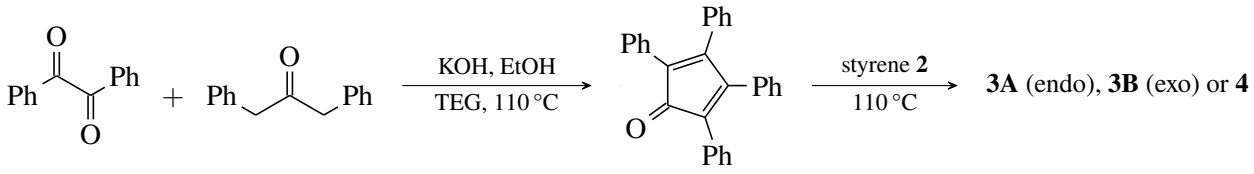


Figure 1: One-pot sequence studied. Tetracyclone **1** (MW 384.48) reacts with styrene **2** to give the bridged ketone adducts **3A/3B** (1,2,3,4,5-pentaphenylbicyclo[2.2.1]hept-2-en-7-one, MW 488.63), which could also lose CO to give the cyclohexadiene **4** (1,2,3,4,5-pentaphenylcyclohexa-1,3-diene, MW 460.62). Structures as drawn in the laboratory manual.[3]

Diels–Alder reaction. The Diels–Alder reaction, first reported in 1928,[4] is a concerted [4+2] cycloaddition between a conjugated diene and an alkene (the dienophile). Two new C–C σ bonds are formed at the expense of two π bonds, so the reaction is thermodynamically favourable. Because all bonds form in a single step without rotation, the reaction is stereospecific.[1,5] Tetracyclone is a good diene because its diene unit is locked in the required s-*cis* geometry within the ring. Styrene is a modest dienophile: it has no strong electron-withdrawing group, but its phenyl substituent activates the double bond sufficiently at 110 °C.[1]

With a cyclic diene and an unsymmetrical dienophile, two diastereomeric bicyclic adducts can form. In the *endo* adduct **3A**, the phenyl group from styrene points towards the longer (two-carbon) bridge; in the *exo* adduct **3B**, it points away from it. Under kinetic control (lower temperature), the endo adduct usually forms faster, whereas the exo adduct is often the more stable, thermodynamic product and can dominate at higher temperature.[1] In addition, adducts of tetracyclone contain a strained bridging carbonyl group that can be lost as carbon monoxide. This happens, for example, with benzyne and with diphenylacetylene, where CO extrusion gives aromatic products.[1,6] With styrene, loss of CO from **3A/3B** would give the cyclohexadiene **4**. Each of the three candidates would be formed as a racemate.

Structure determination. All three candidates contain three non-aromatic hydrogens (one CH and one diastereotopic CH_2), each expected as a doublet of doublets. The geminal coupling is large (11–14 Hz), while the vicinal couplings depend on the H–C–C–H dihedral angle according to the Karplus relationship.[3,7] IR spectroscopy indicates whether the bridging ketone is still present. The stereochemistry, however, is established unequivocally by the X-ray crystal structure, as in the original description of this experiment.[8]

Solvent. Triethylene glycol (TEG) has a very low vapour pressure and a high boiling point (287 °C), tolerates sustained heating above 110 °C and has low toxicity, which makes it suitable for running both steps in one heated vial.[1]

The aims of this experiment were (i) to carry out the one-pot aldol condensation and Diels–Alder reaction, monitoring each step by TLC, (ii) to purify the product by recrystallisation, and (iii) to determine which of **3A**, **3B** and **4** was formed, using the melting point, IR and ^{1}H NMR data and the X-ray crystal structure.

2. Experimental

2.1 General

Benzil, 1,3-diphenylacetone, triethylene glycol, styrene (stabilised with 4-*tert*-butylcatechol), freshly prepared 2 M ethanolic KOH, ethanol, ethyl acetate and hexane were used as supplied. TLC was performed with 10% ethyl acetate in hexane as the eluent, and plates were visualised under UV light. The melting point was measured on a Mel-Temp apparatus. IR spectra were recorded on [instrument] with an ATR accessory. ^{1}H NMR spectra were recorded at [400] MHz in $CDCl_3$; chemical shifts are reported relative to residual $CHCl_3$ (δ 7.26).[9]

2.2 One-pot synthesis

Benzil ([0.210] g, [1.00] mmol) and 1,3-diphenylacetone ([0.210] g, [1.00] mmol) were placed in a 20 mL reaction vial with a small stirrer bar, and triethylene glycol (0.50 mL) was added. The vial was heated on a heating block at 110 °C until the solids had melted or dissolved to give a bright yellow solution. Ethanolic KOH (2 M, [5–6] drops) was added; the mixture turned [purple] and [solidified], and heating was continued for 3–5 min. A small sample (R1) was dissolved in ethyl acetate and analysed by TLC against the two starting materials, which showed [complete] consumption of the starting materials.

Styrene (0.40 mL, 3.48 mmol, 3.5 equiv) was added, and the mixture was heated at 110 °C for 30 min, during which its colour changed from dark purple to [tan]. TLC of a sample (R2) against R1 showed [no 1 left].

2.3 Work-up and recrystallisation

The mixture was cooled to room temperature, ethanol (5 mL) was added and the mixture was stirred with a spatula and cooled in an ice bath. The precipitate was collected by suction filtration on a Hirsch funnel, broken up with a spatula and washed well with ethanol to remove residual styrene. The crude solid was heated to a

gentle boil in a mixture of ethanol (10 mL) and ethyl acetate (10 mL), and further hot ethyl acetate (X mL) was added dropwise until the solid had dissolved. The solution was cooled to room temperature over 15–20 min and then in an ice bath. The crystals were collected by suction filtration, rinsed with a minimal volume of ethanol and dried on the filter to give the product as appearance (0.XXX g, X.XXX mmol, XX %).

Characterisation data. Mp XXX–XXX °C. IR (ATR) $\tilde{\nu}_{\max}$/1/cm: values. ^{1}H NMR (400 MHz, $CDCl_3$) δ: values.

3. Results

Table 1: Quantities of reagents (planned values from the laboratory manual[3] and actual values used).

Reagent	M / g mol^{-1}	Density	Planned	Actual	n / mmol	Equiv.
Benzil	210.23	—	0.21 g	□	1.00	1.00
1,3-Diphenylacetone	210.27	—	0.21 g	□	1.00	1.00
Triethylene glycol	150.17	1.113	0.50 mL	□	solvent	—
KOH (2 M in EtOH)	56.10	—	5–6 drops	□ drops	cat.	—
Styrene	104.15	0.906	0.40 mL	□	3.48	3.5

Table 2: Yield. Keep the row that matches the structure identified.

	If product is **3A/3B** (MW 488.63)	If product is **4** (MW 460.62)
Theoretical yield (from 1.00 mmol benzil)	0.488 g	0.460 g
Mass of recrystallised product	0.XXX g	
Isolated yield	XX %	XX %

Table 3: TLC data (silica, 10% EtOAc/hexane, UV visualisation; solvent front X.X cm).

Sample	Distance / cm	R_f	Comment
Benzil	□	□	
1,3-Diphenylacetone	□	□	
R1 (after aldol step)	□	□	OBS: purple spot = 1?
R2 (after Diels–Alder step)	□	□	OBS: is the spot of 1 gone?

Table 4: Observations.

Stage	Observation
Solids in TEG at 110 °C	OBS: melting/dissolution, bright yellow?
Addition of KOH	OBS: colour change, solidification, number of drops
Addition of styrene, 30 min	OBS: purple to tan? time taken
Ethanol addition and cooling	OBS: precipitate
Recrystallisation	OBS: volume of EtOAc added, crystal form
Melting point	OBS: range; any gas evolution or colour change on heating

Table 5: Diagnostic spectroscopic data of the product.

Measurement	Result
IR, 1650–1850 1/cm region	C=O band at X cm^{-1} / no C=O band
^{1}H NMR, aromatic region	δ range, integral X (25H expected for all candidates)
^{1}H NMR, H_a	δ, dd, J = X, X Hz, 1H
^{1}H NMR, H_b	δ, dd, J = X, X Hz, 1H
^{1}H NMR, H_c	δ, dd, J = X, X Hz, 1H
X-ray crystal structure (provided)	3A / 3B / 4

4. Discussion

4.1 The aldol step

The rapid change from bright yellow to deep purple on addition of KOH shows the formation of the highly conjugated tetracyclone **1**. TLC of R1 confirmed R1 TLC result. Only a catalytic amount of base is needed because hydroxide is regenerated in each dehydration step. Heating in TEG keeps the reactants molten or dissolved without solvent loss, and the product precipitates as the reaction proceeds, which helps to drive it to completion.

4.2 The Diels–Alder step

The fading of the purple colour to tan during heating with styrene indicates that the conjugated cyclopentadienone chromophore of **1** was consumed. In the adducts **3A/3B**, the former diene carbons become sp^3 bridgeheads, so the conjugation is broken. Styrene was used in excess (3.5 equiv) because it is only a moderately reactive dienophile and some may evaporate or polymerise at 110 °C; the excess was removed by washing the crude solid with ethanol.

4.3 Identification of the product

Bridging ketone or not. Compounds **3A** and **3B** contain a carbonyl group bridging a bicyclo[2.2.1] framework. This carbonyl is strongly strained, and ring strain shifts C=O stretching bands to higher wavenumber (for comparison, cyclopentanone absorbs at a higher wavenumber than cyclohexanone).[10] A C=O band well above that of an unstrained ketone is therefore expected for **3A/3B**, whereas **4** has no carbonyl group at all. The IR spectrum showed C=O at X cm^{-1} / none, which supports 3A/3B / supports 4.

Non-aromatic protons. All three candidates have 25 aromatic and 3 non-aromatic protons, so the integral ratio (25.0 : 3.0) confirms that a 1:1 adduct was formed but cannot distinguish between them. The three non-aromatic signals appeared as doublets of doublets. The largest coupling, X Hz, lies in the 11–14 Hz range expected for the geminal pair of CH_2 protons.[3] The vicinal couplings between the benzylic CH and the two CH_2 protons follow the Karplus relationship:[7] couplings are largest for dihedral angles near 0° or 180° and smallest near 90°. In the rigid bicyclic skeleton of **3A/3B**, the hydrogens are held in eclipsed positions, so the benzylic H is *cis* to one CH_2 proton (dihedral angle close to 0°, large J) and *trans* to the other (larger dihedral angle, smaller J). In the flexible cyclohexadiene ring of **4**, the dihedral angles are instead averaged over ring conformations.

Interpret your two vicinal J values here (which CH_2 proton is cis and which is trans to the benzylic H).

X-ray structure. The crystal structure provided showed 3A / 3B / 4, in agreement with the spectroscopic interpretation. This illustrates that IR and NMR spectroscopy can narrow down the possibilities, but an X-ray

structure is needed to establish the relative stereochemistry unequivocally.[8]

Melting point. The product melted at XXX–XXX °C. The laboratory manual notes that the product may release styrene when heated above 170 °C,[3] that is, it may undergo a retro-Diels–Alder reaction.

Comment on any gas evolution or colour change observed near the melting point.

4.4 Yield and efficiency

The isolated yield was XX %.

Discuss losses: material left on the vial and Hirsch funnel, solubility in the ethanol washes and in the recrystallisation mother liquor, incomplete reaction (TLC of R2).

The sequence is highly atom-economical. The aldol step loses only two molecules of water, giving an atom economy of $384.48/(210.23 + 210.27) = 91.4\%$ for **1**; the Diels–Alder step itself incorporates all atoms of both partners. The overall atom economy from benzil, 1,3-diphenylacetone and styrene is 93.1% for **3A/3B**, or 87.8% for **4** if CO is lost.[11] Running both steps in one vessel also avoided isolating, drying and redissolving tetracyclone.

5. Conclusion

Tetracyclone was prepared by a double aldol condensation and trapped in situ with styrene in a one-pot sequence in triethylene glycol. The product was obtained in XX % yield after recrystallisation and identified as 3A / 3B / 4 on the basis of its IR and ^{1}H NMR spectra and the X-ray crystal structure. The experiment shows how a one-pot sequence of two atom-economical reactions can build a complex, polyaryl bicyclic framework, and why X-ray crystallography is needed to confirm stereochemistry that spectroscopy alone can only suggest.

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